Evaluating performance of fine-tuned benchmark CNNs and geometry encoded MLP for sparse crowd counting applications

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Project Statement

- What is the problem?
  - Current limitations of transfer learning
    - Poor generalization on various application
    - Domain shift usually occurs with various lighting, camera angle and geometric distortion that impact model accuracy
  - Lack of studies encode data geometry as inductive bias with image processing
- Our objective
  - Practicing the transfer learning method on popular crowd-counting model to meet specific application goals
  - Use image processing and implicit system geometry to highlight human readable inputs

Motivation

- Why is this important?
  - Training a functional crowd counting model using limited data, save time and effort
  - Provides a guideline for future image based projects
  - Compares benchmarks ability across domains and highlights meaningful preprocessing
- Real-world relevance
  - Estimating attendance in campus events
  - Provide data driven insight to events planning

Background

- Key concepts
  - Timeline of release of the models

2000-2012

Detection-based methods before deep learning.

2018

CSRNet: Dilated convolutional neural networks for understanding the highly congested scenes.

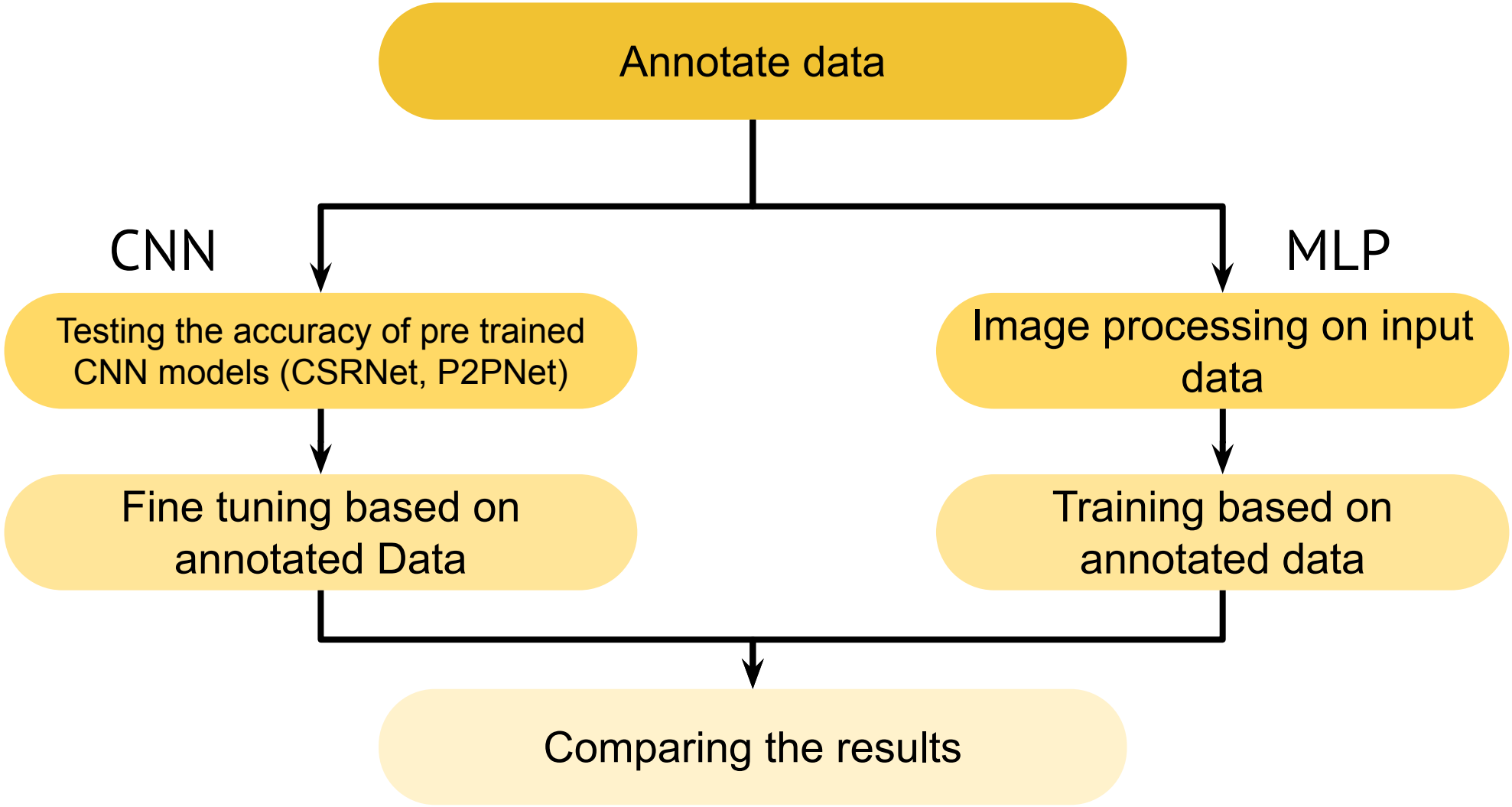
2021

P2PNet: Rethinking counting and localization in crowds: A purely point-based framework.

2024

Proposal: identifying geometric structure in data and how leveraging data geometry can give rise to efficient ML algorithms with provable guarantees.

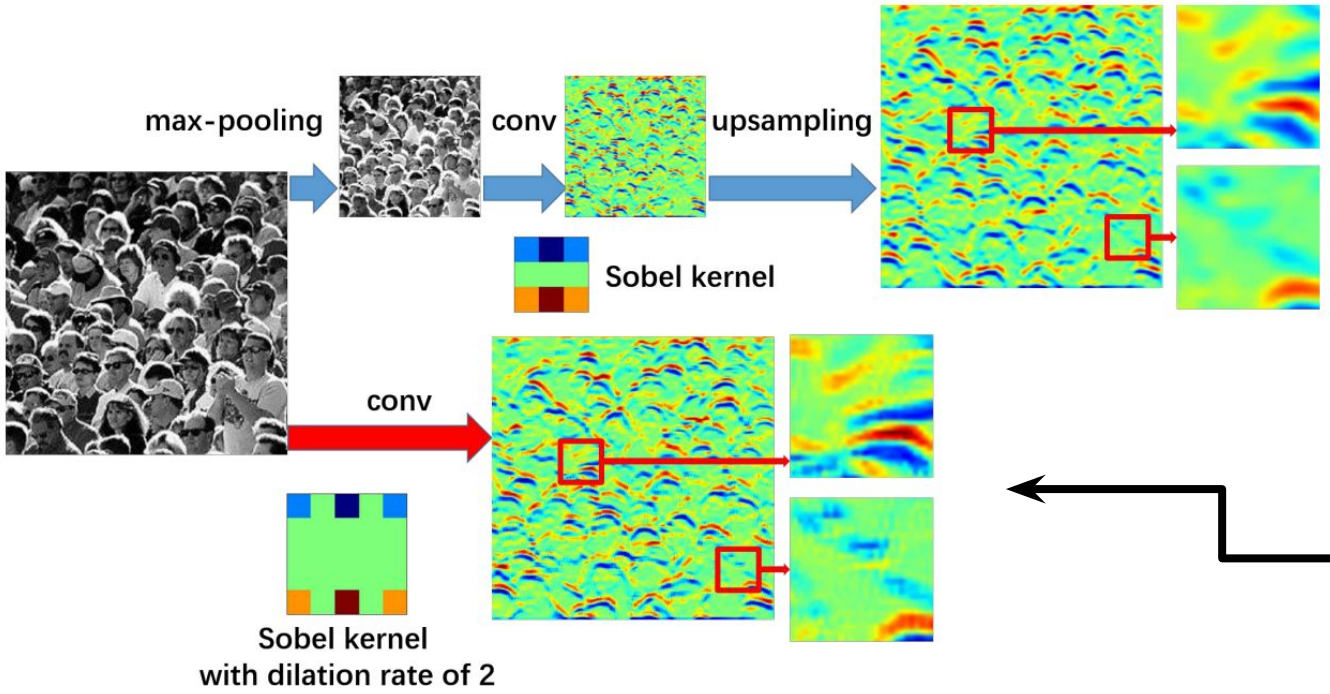
Method Overview



Data Annotation

Defines the ground truth coordinates/count for people in the frame

Model Details - CSRNet



Input: Unfixed-resolution color image

↓

Front-end: Fine-tuned VGG16

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Back-end: Fully connected Dilated Convolutional layers

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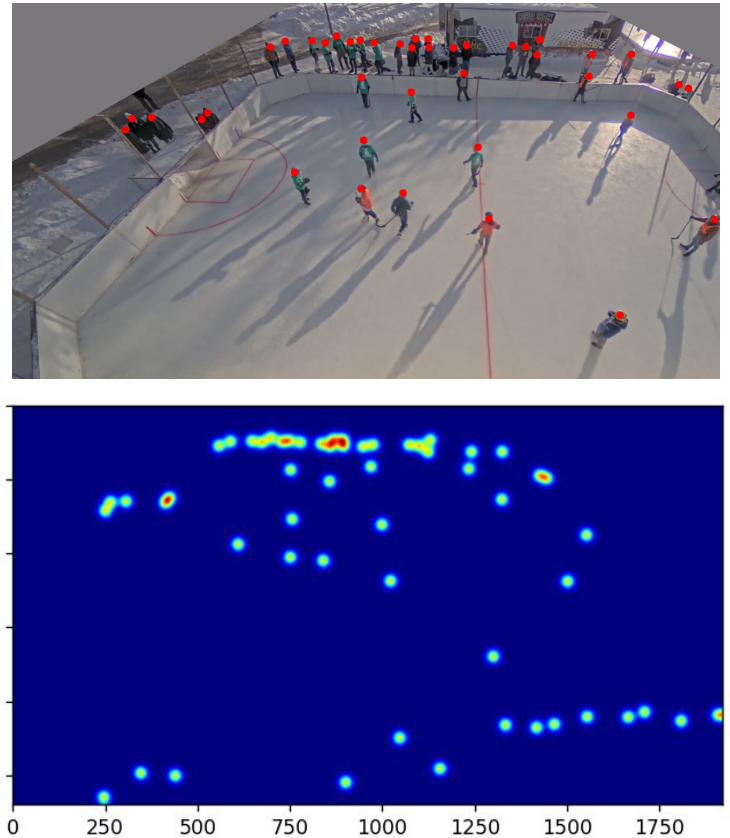
Density Map: conv1-1-1

- Dilated convolutional layers preserve spatial resolution
- Enables effective estimation of dense crowd distributions without additional pooling operations

Workflow- CSRNet

1. Convert the annotated coordinates to density heatmap using gaussian filter.
2. Load the pre-trained weights and frozen the front-end layers to preserve the low level features.
3. Train only the back-end layers with dilation rate of 2.



Model Details + Workflow - P2PNet

- Algorithm used
  - One-to-one Hungarian matching
- Key layers
  - Backbone(VGG 16) feature extraction
  - FPN-based muti-scale fusion
  - Two predicted heads
- Modifications
  - compared original pre-trained model vs fine-tuned model
- Workflow
  - Input image → VGG backbone feature extraction → muti-scal fusion → prediction heads for point matching → counting and approximate localization of each people

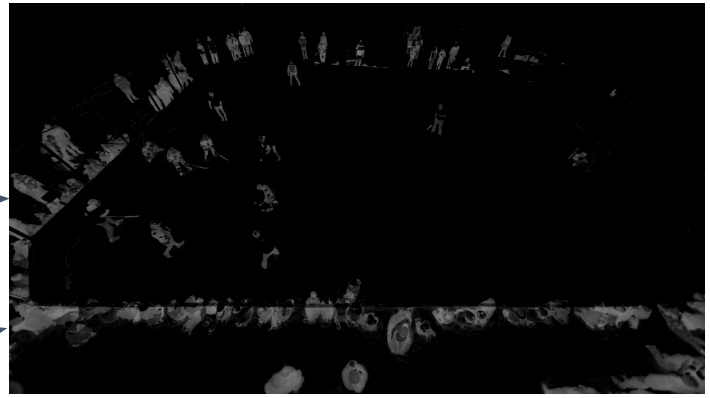
Preprocessing Filter - MLP

Processes:

- When subtracting affixed background:
- We did subtracted input from background, better experimentally
- We averaged the HSV value before subtraction to match the background to our test

1a. Example Input Image

1b. Example Background Image



2. Background Subtraction Grayscale, Average Background

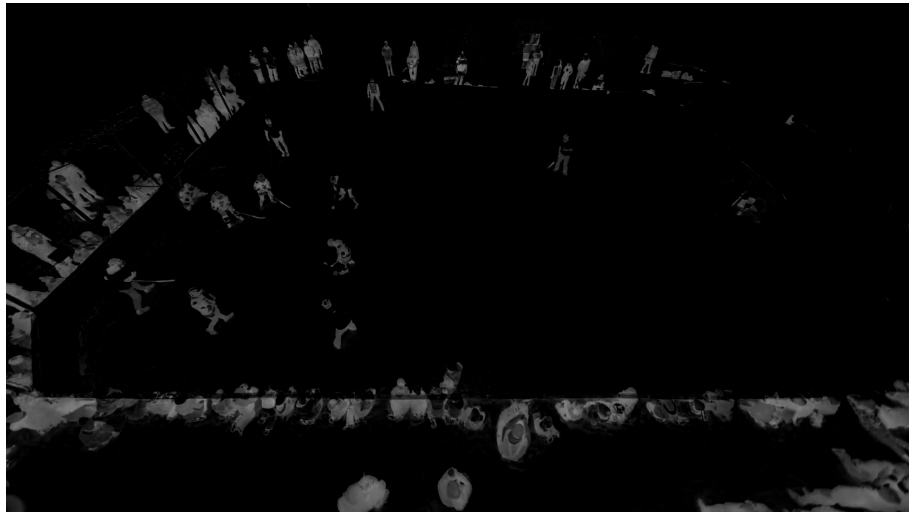
Preprocessing Contour- MLP

Processes:

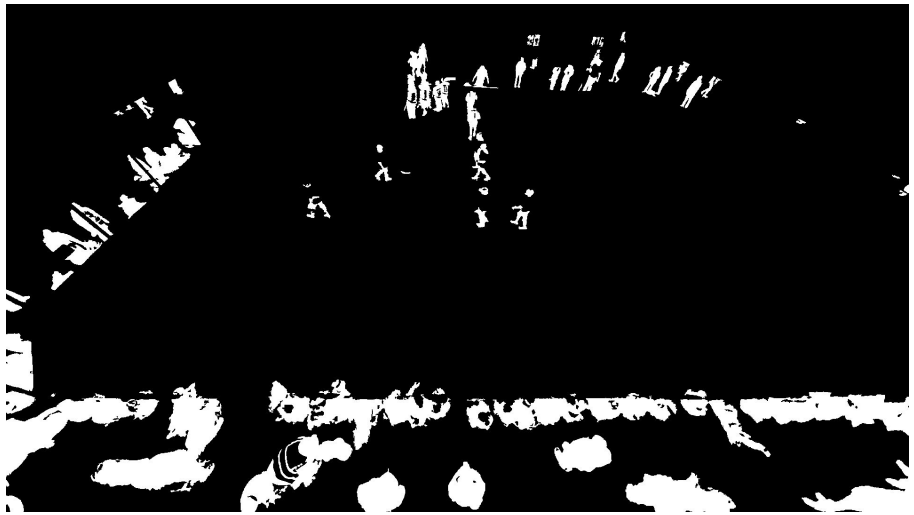
- Used OpenCV for contours and dilation,
- Contour global threshold for 20-255
- Then dilated mask with a 5x5 kernel

Notes on output:

- Annotations not optimized for digital processing
- Weirdness with orange jerseys causing heads to float



2. Background Subtraction Grayscale, Averageback ground



3. Dilated Contour mask of previous Grayscale

Model Details Inputs - MLP

- Trained on each individual contour found
  - 6557 total data instances, compared to 100 images
  - Five input parameters from OpenCV
    - Radius from pixel below camera (pixel)
    - Theta from pixel below camera (radians)
    - Normalized Dot product to radius tangent
    - Area (pixels)
    - Solidity (%)

Same zoom comparison to close children vs. far adults

Model Details + Workflow - MLP- continue

Algorithm:

- Key parameters
  - Adam Optimizer- Start LR is 0.005
  - L2 Regularization- L2 Weight is 0.01
  - 60 Epochs, taking best model
- Potential Improvements/Test-
  - Induce symmetry across theta
  - Run with dataset on any part of a person seen visually

Layer (type)	Output Shape	Param #
input_layer_50 (InputLayer)	(None, 5)	0
dense_236 (Dense)	(None, 256)	1,536
dense_237 (Dense)	(None, 256)	65,792
dense_238 (Dense)	(None, 256)	65,792
batch_normalization_33 (BatchNormalization)	(None, 256)	1,024
activation_33 (Activation)	(None, 256)	0
dropout_53 (Dropout)	(None, 256)	0
dense_239 (Dense)	(None, 1)	257

Total params: 134,401 (525.00 KB)
Trainable params: 133,889 (523.00 KB)
Non-trainable params: 512 (2.00 KB)

- Per blob MSE -
  - Test = 0.538
  - Validate = 0.7097

Experimental Setup

- Dataset: 1 game on 02/05, 10:10 am - 11:00 am, 100 frames with 30 second in between each frame
- Train/Test split: 60, 20, 20
- Metrics (accuracy): Mean Absolute Error (MAE)

Model	Default model MAE (Person)	Fine Tuned MAE (Person)
CSRNet	10.821	12.2
P2PNet	39.4709	5.99
Preprocessed MLP	38.0	24.6

Observations

- CSR overfitting with small dataset
- MLP mask sometimes didn't contain faces
- P2PNet seen a increase in accuracy after fine-tuning

Conclusions

- P2P training was most successful with our small dataset
- The MLP performed near CSRNet preformance
- Preprocessing adds a layer of observability and the ability to increase SNR (Person to background), but requires carefulness

Work left to do for report

- Run second day data annotation
- Improve and troubleshoot data preprocessing
- More precise fine tuning
- Partition training set for MLP to use blobs of only certain frames similar to the other models

Future Work + Q&A

Potential Works:

- Image segmentation filter that spares blobs to images for a CNN input
- Image preprocessing in a controlled environment ML applications

Improvements to our study:

- Larger annotated dataset
- Increase variety of times

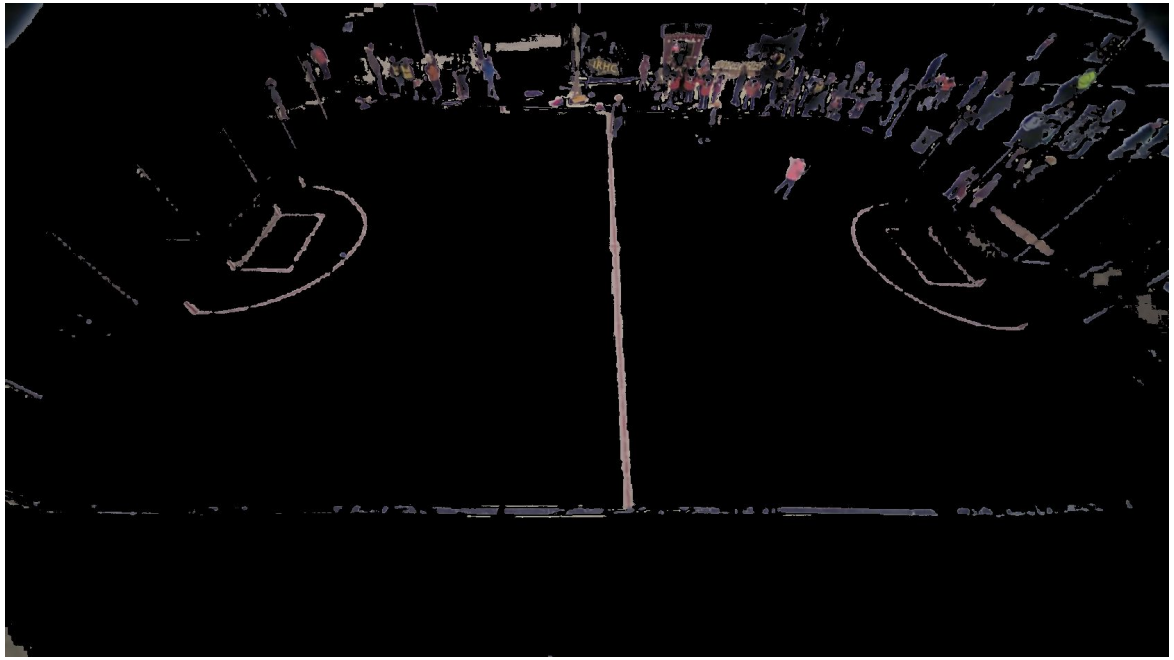
References

[1] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma, “Single-Image Crowd Counting via Multi-Column Convolutional Neural Network,” *IEEE Xplore*, 2016. https://ieeexplore.ieee.org/document/7780439 (accessed Apr. 22, 2021).

[2] Y. Li, X. Zhang, and D. Chen, “CSRNet: Dilated Convolutional Neural Networks for Understanding the Highly Congested Scenes,” *arXiv:1802.10062 [cs]*, Apr. 2018, Accessed: Apr. 22, 2021. [Online]. Available: https://arxiv.org/abs/1802.10062

[3] X. Cao, Z. Wang, Y. Zhao, and F. Su, “Scale Aggregation Network for Accurate and Efficient Crowd Counting,” *Lecture Notes in Computer Science*, pp. 757–773, Jan. 2018, doi: https://doi.org/10.1007/978-3-030-01228-1_45.

[4] M. Weber, “Exploiting Data Geometry in Machine Learning,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 20, pp. 22681–22681, Mar. 2024, doi: https://doi.org/10.1609/aaai.v38i20.30297.

Other Image Processing ways

- Manual snow removal,
- Filter out the color and near colors of snow